

UNIVERSITY OF SINDH

JAMSHORO, SINDH, PAKISTAN



ASSIGNMENT # 01

Human-Computer Interaction & Computer Graphics

Course Code: HCI & CG

SUBMITTED TO:

MR. RAJESH KUMAR
Course Instructor

SUBMITTED BY:

MUHAMMAD YOUSIF
S/O Mehmood Babar

ROLL NUMBER:

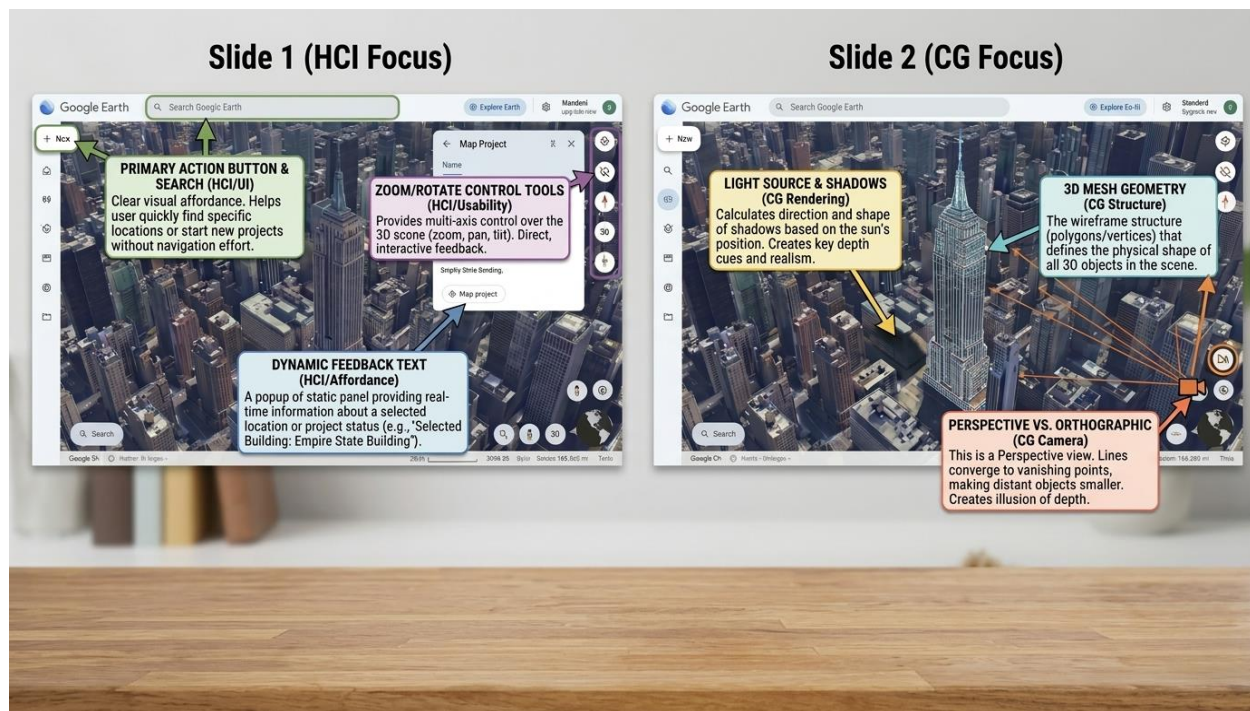
2k24/cse/108

DEPARTMENT:

IMCS
(Institute of Mathematics & Computer Science)

DATE OF SUBMISSION

September 12, 2026



Question 1: How CG Quality Helps/Hinders

- Detailed Explanation:** High Computer Graphics (CG) quality enhances usability by adding visual depth, realistic shadows, and accurate 3D geometry. These visual cues make spatial navigation intuitive because users can easily gauge distances, building shapes, and terrain features as if looking at a real-world object. However, rendering complex 3D meshes and dynamic real-time lighting places a heavy computational load on the GPU and CPU. On lower-end devices or slow network connections, high CG quality can cause lag, dropped frame rates, and delayed UI responsiveness, ultimately hindering the user experience.

Question 2: Simple HCI Change

- Detailed Explanation:** To improve usability under Human-Computer Interaction (HCI) principles, adding a dedicated **"3D View Toggle"** button directly to the primary toolbar reduces navigation friction. Currently, switching between 2D map views and 3D globe perspectives requires multi-step interaction or dynamic gesture controls. Placing a prominent toggle switch in the main interface gives users immediate visual affordance, allowing them to instantly switch modes based on their device performance or task needs without searching through nested menus.

